Web GIS Project

Different methods for visualization of US wildfire (2005, 2010, 2015, 2020)

And relation with NDVI

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Introduction:

Unplanned, uncontrolled, and unpredicted fire in an area of combustible vegetation is known as wildfires. Wildfires has mainly two causes humans and nature. Nearly 85% of wildfires in United States are caused by humans. Humans-caused fires result from campfires left unattended, the burning of debris, and intentional acts of arson. In natural cause, there comes lightning. Fires are usually started by unusually long-lasting hot lightning bolts.

Beginning in the mid-1980s, forest wildfire activity in US forests were due to climatic factors such as warmer temperatures, dry summers, below average winter precipitation or earlier spring snowmelt.

So this work is done to visualize or represent the geospatial data by three different methods,

I. Through QGIS
II. On Web by using leaflet (using GeoJSON file)
III. On Web by using leaflet (with using GeoServer)

As shows in below fig the Acres burnt in America,

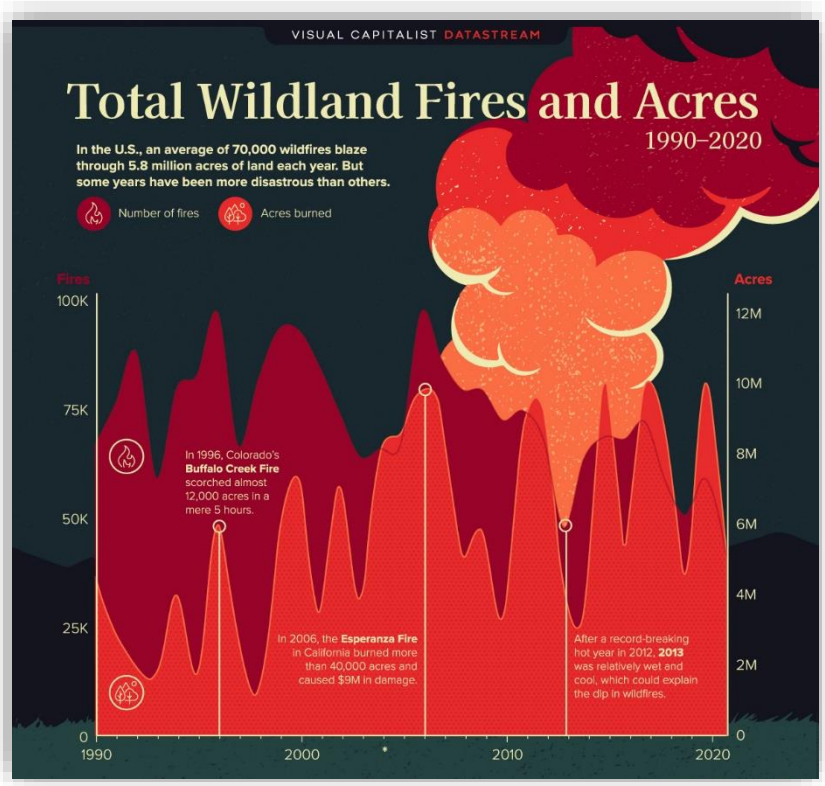


Figure 1 shows the relation between fires and area burnt in America

Data Source:

The data is collected form the [United States Department of Agriculture Forest Services](). Here I download ESRI geodatabase. I download the Wildfires data for year 2005, 2010, 2015, and 2020.

Data Description:

The shape file contains the intensity of places where wildfire is occurred in the conterminous United States (CONUS), Alaska, Hawaii, and Puerto Rico from the beginning of Thematic Mapper archive to the present. Downloaded data contains type of fires, ACRES burnt and other as shown below,

FIRE_TYPE	IG_DATE	LATITUDE	LONGITUI	ACRES	MAP_ID	MAP_PROG
Wildfire	19/06/05	34.177	-116.502	2004	14129	MTBS
Wildfire	20/06/05	33.198	-86.36	748	13826	MTBS
Wildfire	20/06/05	65.608	-147.646	184179	13978	MTBS
Wildfire	20/06/05	34.172	-112.121	6303	14023	MTBS
Wildfire	21/06/05	34.243	-111.877	246143	13883	MTBS
Wildfire	21/06/05	42.993	-115.466	3100	14039	MTBS
Wildfire	21/06/05	42.782	-115.417	1356	14046	MTBS
Wildfire	21/06/05	33.164	-109.193	1721	14196	MTBS
Wildfire	21/06/05	33.723	-111.271	15849	14208	MTBS
Wildfire	21/06/05	42.723	-115.776	1366	14387	MTBS
Wildfire	22/06/05	35.597	-114.48	18064	14025	MTBS
Wildfire	22/06/05	34.049	-116.559	3461	14028	MTBS
Wildfire	22/06/05	42.681	-115.025	5269	14047	MTBS
Wildfire	22/06/05	37.166	-114.619	150019	14076	MTBS
Wildfire	22/06/05	37.242	-114.303	327156	14077	MTBS
Wildfire	22/06/05	36.663	-114.182	2970	14081	MTBS
Wildfire	22/06/05	36.421	-115.014	3876	14082	MTBS
Wildfire	22/06/05	35.917	-115.471	33448	14083	MTBS
Wildfire	22/06/05	39.753	-112.582	1409	14110	MTBS
Wildfire	22/06/05	33.817	-116.918	2031	14147	MTBS
Wildfire	22/06/05	35.768	-120.831	1279	14161	MTBS
Wildfire	22/06/05	37.152	-106.896	1083	14195	MTBS

Figure 2 shows the data download in csv format

Methodology:

1. Through QGIS:

   In this method, I simply upload the shapefile to QGIS, and then use Heatmap Symbology to make a heatmap. In this, I use ACRES as weighted points. After this I get heatmap, of which year 2005,2010,2015, and 2020 are below and other from 1984 to 2020 are on uploaded on [drive]().

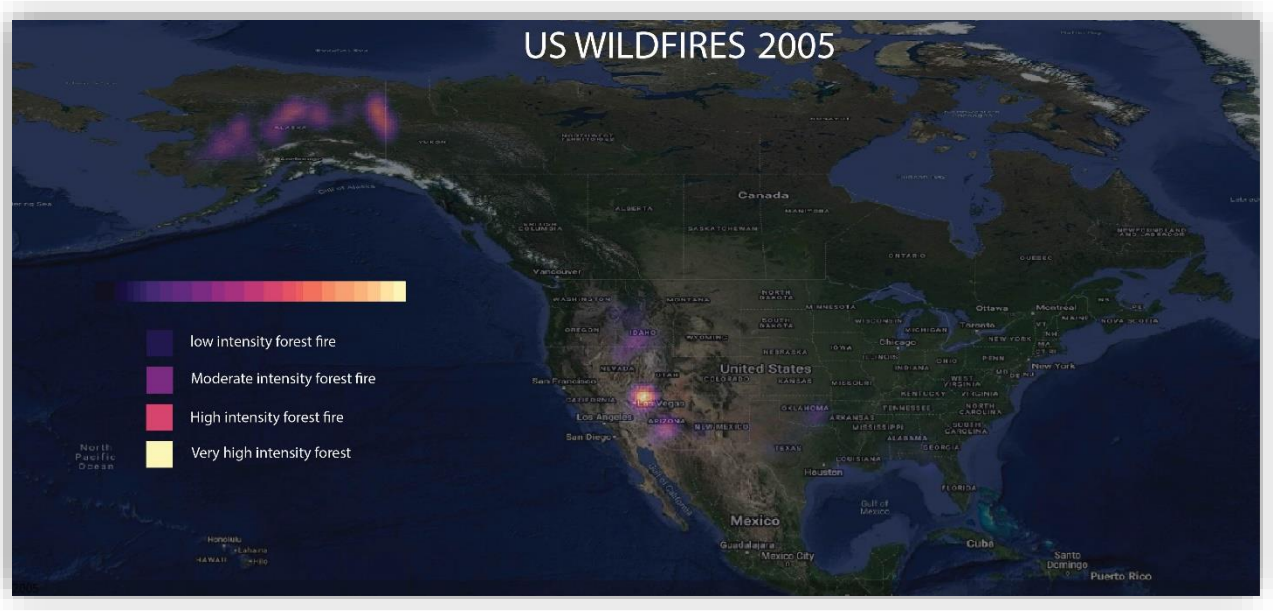


Figure 3 shows heatmap of US 2005 Wildfires through QGIS



Figure 4 shows heatmap of US 2010 Wildfires through QGIS

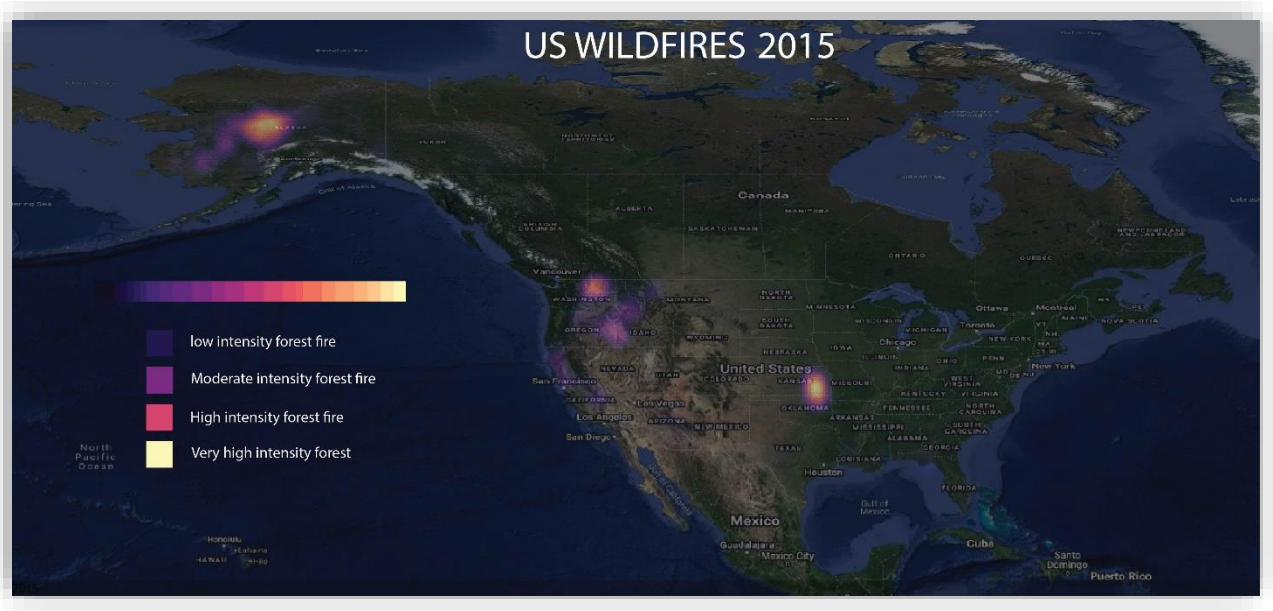


Figure 5 shows heatmap of US 2015 Wildfires through QGIS

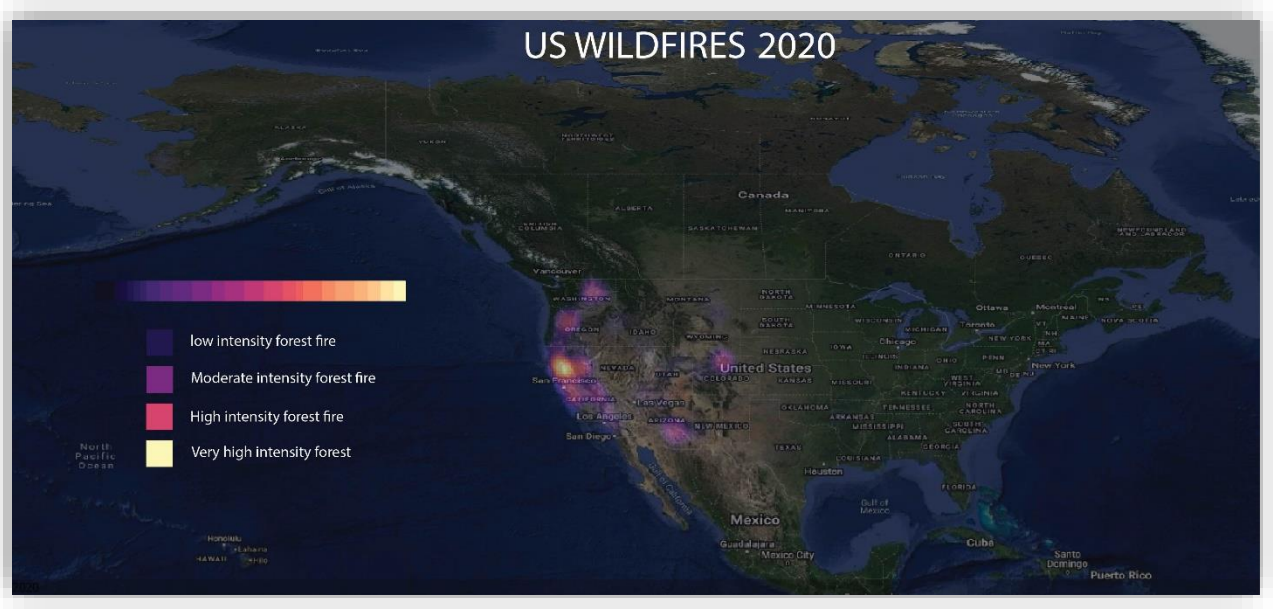


Figure 6 shows heatmap of US 2020 Wildfires through QGIS

2. On Web by using leaflet (using GeoJSON file)

In this method, we have to visualize the data on the web page. For this, we must deal with Html, CSS, and Java-Scripts languages with the help of Leaflet which is a leading open-source JavaScript library for mobile-friendly interactive maps.

Data is uploaded in the form of GeoJSON format which is an open standard geospatial data interchange format that represents simple geographic features and their nonspatial attributes. Results shown or the final Webpage of this method looks like as, which has a slider through which different maps can be visualize,

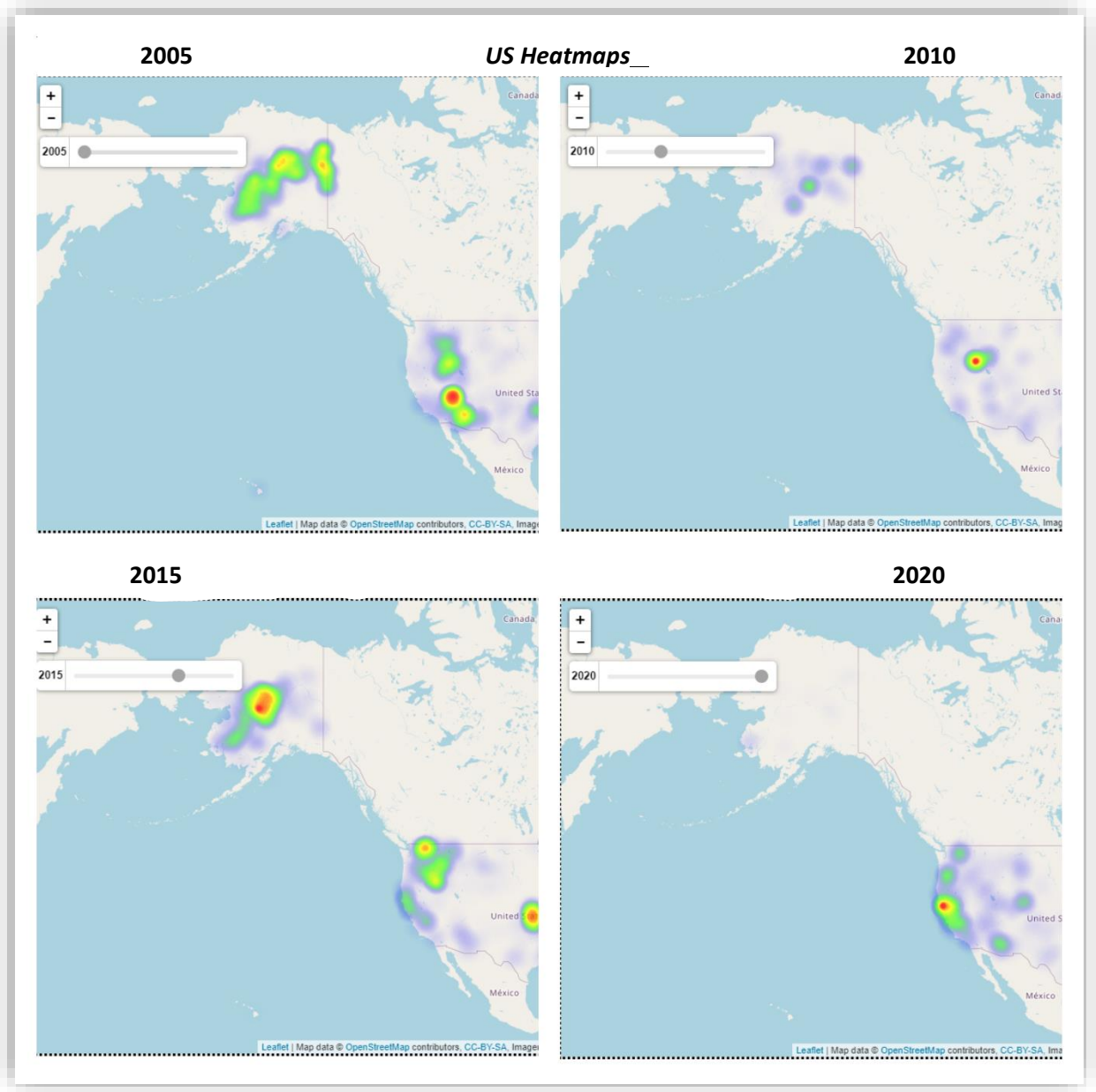


Figure 7 shows 2005, 2010,2015, 2020 US Heatmaps using GeoJSON file with leaflet

3. On Web by using leaflet (with using GeoServer)

In this method, we again have to visualize the data on the web page. But for this, languages like Html, CSS, and Java-Scripts with Leaflet we have to combine the GeoServer, which is an open-source server written in Java that allows users to share, process and edit geospatial data.

First the data (SHP file) is uploaded on the GeoServer, there Heatmap style is required which is taken from the website. After this it can be visualize online using OpenLayers, which is like leaflet needed to visualize the map.

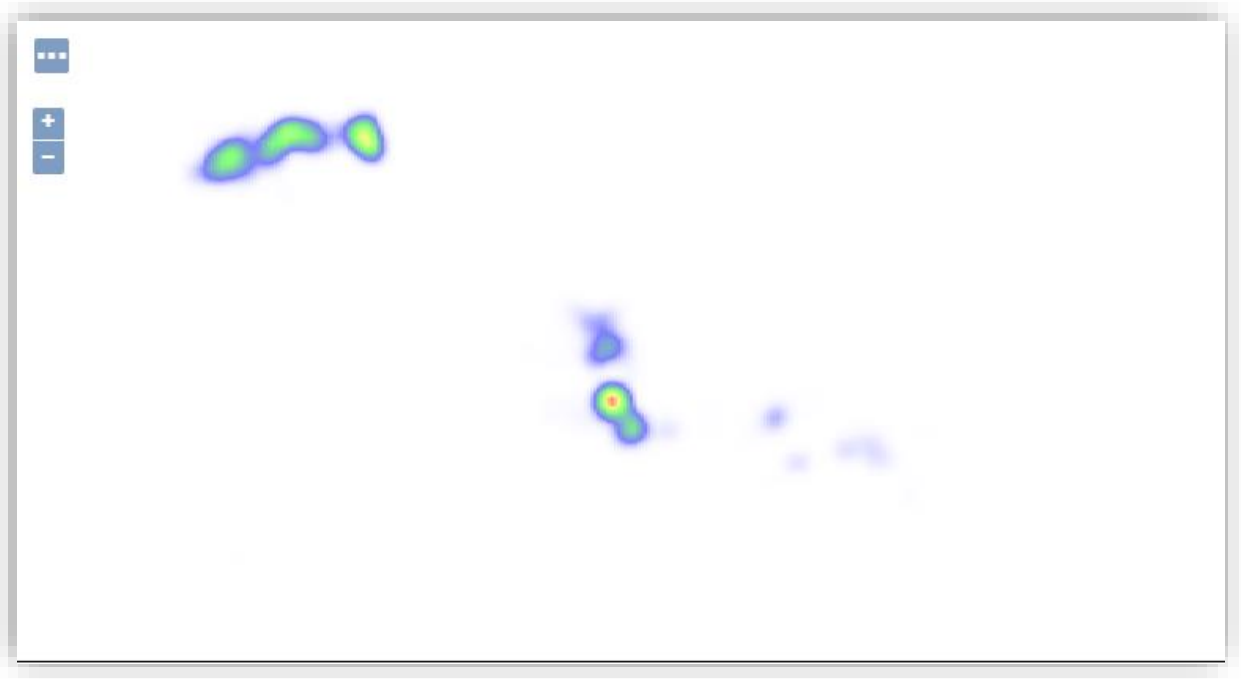


Figure 8 shows heatmap of 2005 US wildfires on GeoServer

Then to add it on the Webpage, Web languages is required. After that results are shown below,

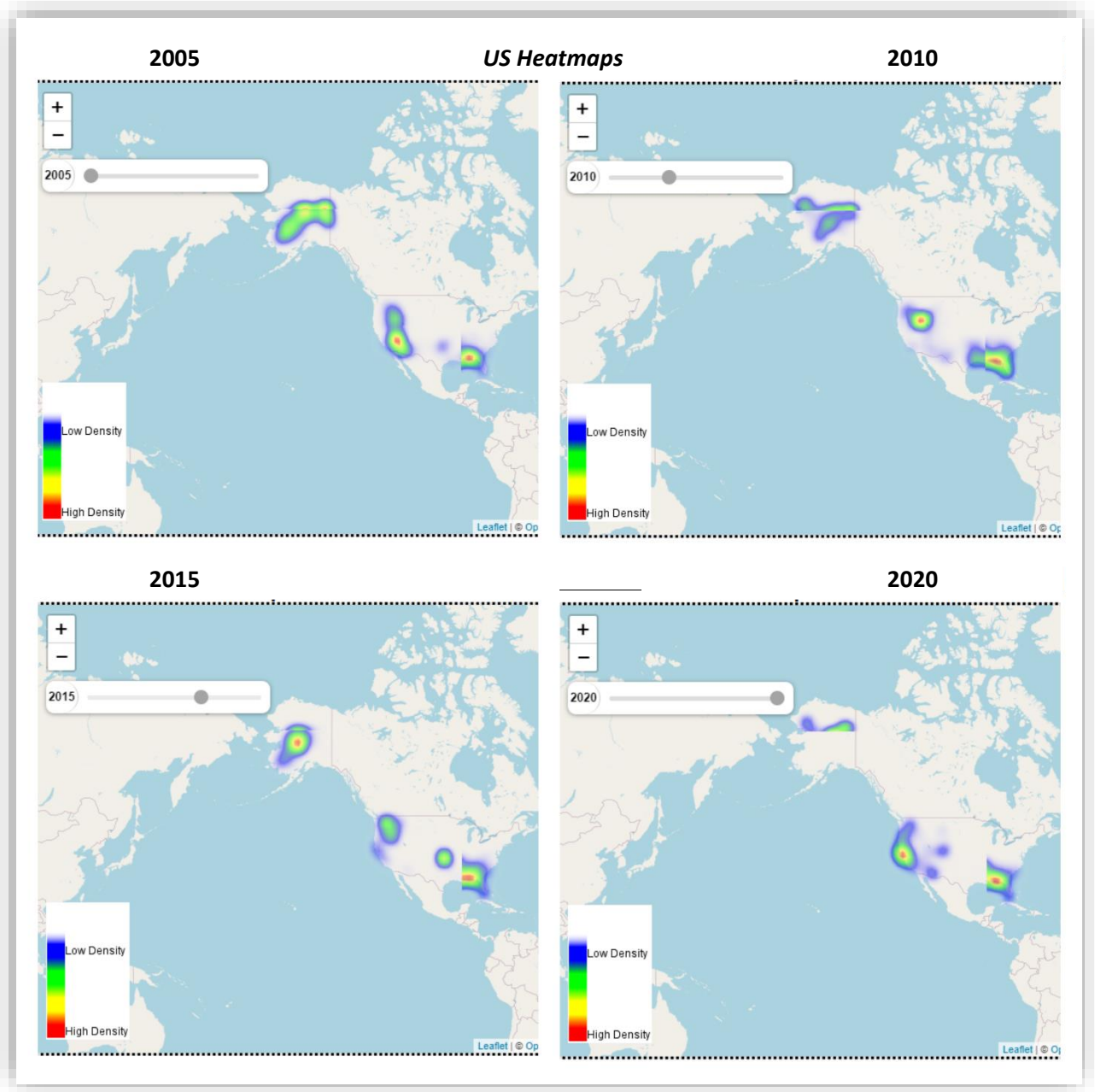
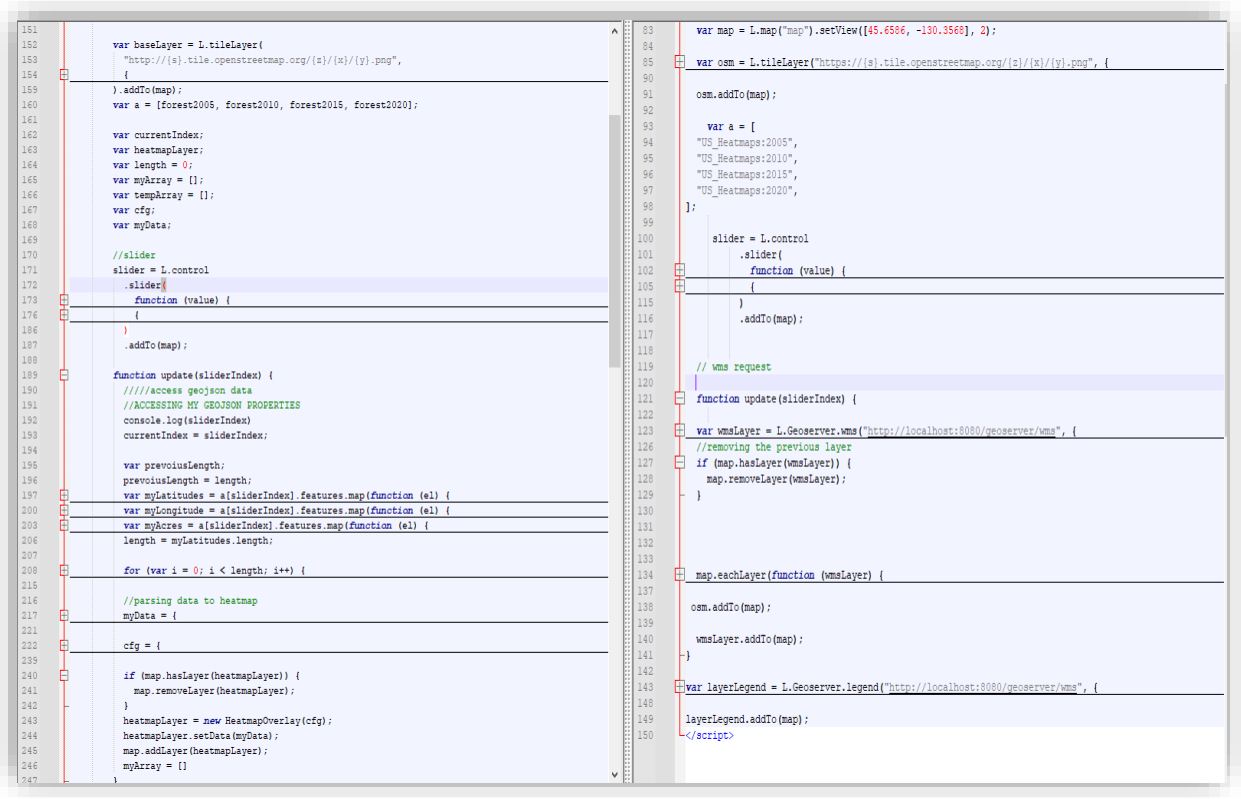


Figure 9 shows 2005, 2010,2015, 2020 US Heatmaps using GeoServer with leaflet

Codes for the 2nd and 3rd methods are uploaded on drive.

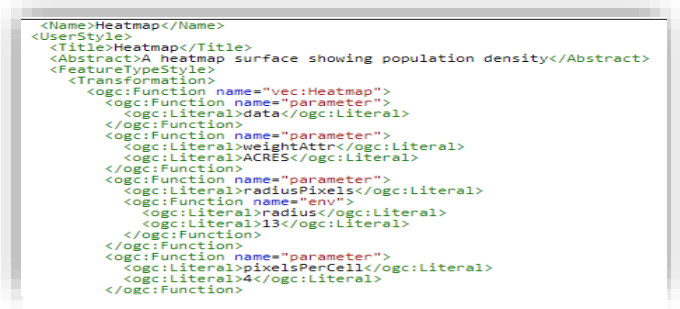
Code Comparison of method 2nd and 3rd:

Codes used to make maps for 2nd and 3rd method visualization of geospatial data is uploaded on drive but here is the comparison of their JavaScript's:



*Left side of **Figure 10** for 2nd method while right side for 3rd method*

In left side of Figure (10) first we put the base map and sets its parameters. Then using update function to call the latitude, longitude, and weighted value (ACRES) from the JS files and stored in array, which then be sent to the *HeatmapOverlay* function which is defined by the leaflet, and we get our desired results as in Figure (7). In the right side of Figure (10) we also put base map and set its parameters, but then we have to call the layer or layers, which we want to show. In this method, instead of using the leaflet Heatmap function we use the code as in Figure (11) to make a heatmap layer which is then called in html by using *L.Geoserver.wms* function and legend by *L.Geoserver.legend.*



```
<Name>Heatmap</Name>
<UserStyle>
  <Title>Heatmap</Title>
  <Abstract>A heatmap surface showing population density</Abstract>
  <FeatureTypeStyle>
    <Transformation>
      <ogc:Function name="vec:Heatmap">
        <ogc:Function name="parameter">
          <ogc:Literal>data</ogc:Literal>
        </ogc:Function>
        <ogc:Function name="parameter">
          <ogc:Literal>weightAttr</ogc:Literal>
          <ogc:Literal>ACRES</ogc:Literal>
        </ogc:Function>
        <ogc:Function name="parameter">
          <ogc:Literal>radiusPixels</ogc:Literal>
          <ogc:Function name="env">
            <ogc:Literal>radius</ogc:Literal>
            <ogc:Literal>13</ogc:Literal>
          </ogc:Function>
        </ogc:Function>
        <ogc:Function name="parameter">
          <ogc:Literal>pixelsPerCell</ogc:Literal>
          <ogc:Literal>4</ogc:Literal>
        </ogc:Function>
```

Figure 11 part of heatmap styling code on GeoServer

NDVI Comparison:

Here is the comparison for the NDVI of US, which is taken the AppEEARS. Here I upload the shapefiles of my data and it gives me the graph as:

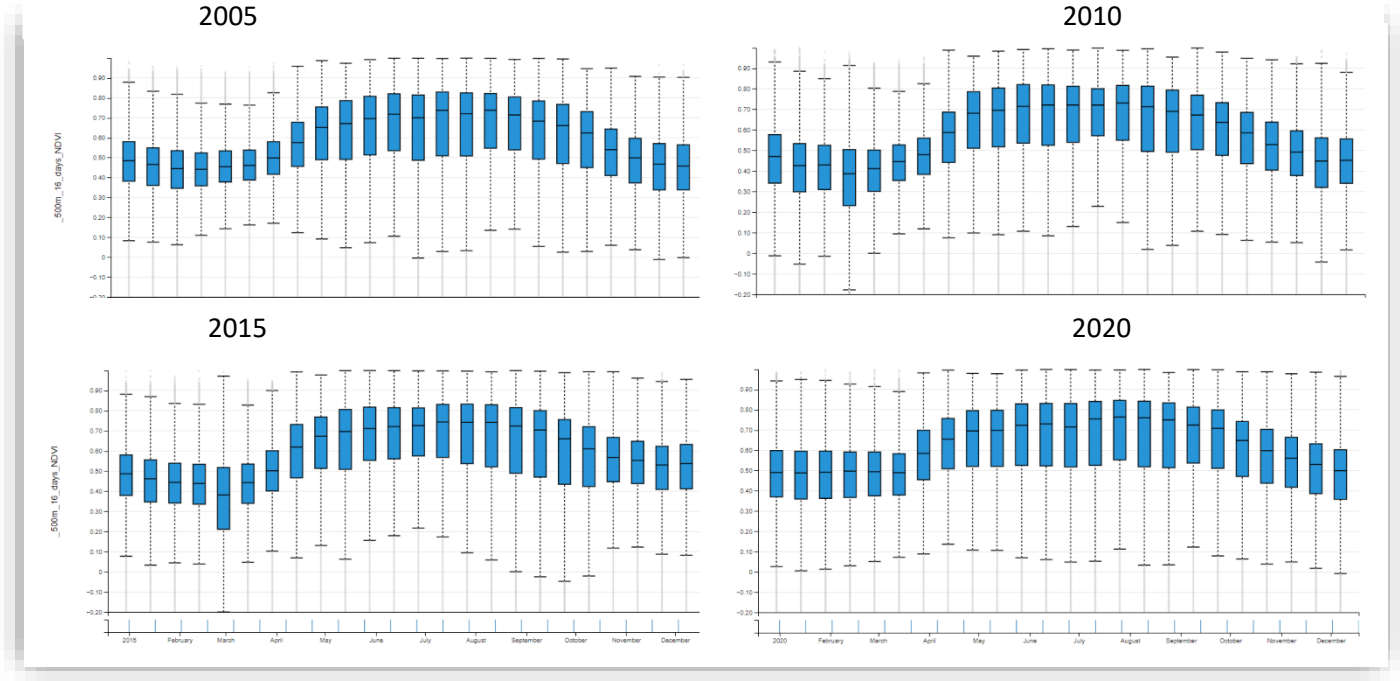


Figure 12 shows the NDVI of 2005, 2010, 2015, 2020 of US

Conclusion:

From the time series mapping, we see that there are specific places where wildfire occurred rapidly which may depend upon the climate conditions like increasing the humidity level. Figs (12) clearly shows that the vegetation increases in the middle of year like from mid of March to mid of July and then goes down, which tells that the in this period (from March to July) there is no or less wildfires.

The difference between the maps of these three methods are due to the difference in values of

- Radius pixels values
- Pixels per cells
- Opacity

And the distortion in the maps of GeoServer are due to the old libraries used in html which may not be compatible with the latest version of GeoServer. So, with the use of GeoServer, it became easy to call our data without taking account how much is its size. But to visualize we must start the GeoServer prior.

References:

- https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4874415/
- https://www.nps.gov/articles/wildfire-causes-and-evaluation.htm#:~:text=Nearly%2085%20percent*%20of%20wildland,and%20intentional%20acts%20of%20arson.&text=Lightning%20is%20one%20of%20the%20two%20natural%20causes%20of%20fires.
- https://www.visualcapitalist.com/30-years-of-wildfires-in-america/
- https://doc.arcgis.com/en/arcgis-online/reference/geojson.htm#:~:text=GeoJSON%20is%20an%20open%20standard,variety%20of%20geographic%20data%20structures.
- https://docs.geoserver.org/2.22.x/en/user/styling/sld/extensions/rendering-transform.html